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Apparatus and method for clock frequency estimation with least squares method

Abstract

An apparatus includes a clock signal input, a sampling circuit, and an estimation circuit. The clock signal input receives N time measurements. A time measurement denotes a respective portion of a given cycle of a clock signal. The sampling circuit is to generate a first sampled window from the clock signal input. The first sampled window includes an accumulation of the N time measurements. The sampling circuit is to generate a second sampled window from the clock signal input including an accumulation of a plurality of products. The estimation circuit is to estimate the frequency or period of the clock signal based upon the first sampled window and the second sampled window.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. provisional application Ser. No. 63/604,992, filed on Dec. 1, 2023, the disclosure of which is incorporated by reference in its entirety for all purposes.

FIELD OF THE INVENTION

(1) The present disclosure relates to clock signals of electronic devices and, more particularly, to an apparatus and method for clock frequency estimation with a least squares method.

BACKGROUND

(2) Clock frequency estimation is used in many electronic systems. Accurate clock frequency estimation seeks to minimize the error involved in such estimations. The accuracy of the clock estimation may be related to N, the number of clock edges that are observed to estimate the constituent clock frequency. One estimation method may be considered more accurate than another estimation method if the error is statistically reduced between estimation methods by increasing the value of N.

(3) Existing solutions such as the one illustrated in FIG. 1 may implement a brute-force calculation of average clock period T during a sample period T.sub.s as in the following equation:

$$(4) \quad T = \frac{t_{N-1} - t_0}{N} \quad \text{equation 1}$$

where t.sub.N-1 is the time at the end of the sample period, t₀ is the time at the beginning of the sample period, and N is the number of measured clock periods during the sample period. The estimation error, ε, of this calculation may be given as

$$(5) \quad \epsilon^2 = \frac{2\sigma^2}{N^2} \quad \text{equation 2}$$

where each clock rising edge has independent measurement noise with variance of σ^2 . These noises include input clock noise, input clock wander, input clock jitter, sample clock noise, sample clock wander, sample clock jitter, resolution, and inherent limitations of the estimation. The estimation error decreases with increasing measurement window T.sub.s, with its commensurate increase in N. The error reduction is proportional to 1/N.

(6) The clock frequency may be computed as the inverse of clock period T. As such, estimation of a clock period may be equivalent to estimation of a clock frequency. However, such brute-force solutions may involve large and complex circuitry, or much memory or processor time.

(7) Simpler solutions may compute a cycle-by-cycle frequency. However, these may introduce a large error.

(8) Examples of the present disclosure may address one or more of these issues.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is an illustration of a brute-force calculation of average clock period T during a sample period T.sub.s.

(2) FIG. 2 is an illustration of an apparatus for estimating a clock cycle or period according to a least squares method, according to examples of the present disclosure.

(3) FIG. 3 is a more detailed illustration of an apparatus for estimating a clock cycle or period

according to a least squares method, according to examples of the present disclosure.

(4) FIG. 4 illustrates an apparatus for estimating a clock period or frequency using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(5) FIG. 5 is a more detailed illustration of an apparatus for estimating a clock cycle or period using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(6) FIG. 6 is another illustration of operation of an apparatus using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(7) FIG. 7 is an illustration of simulation of error response for various values of delay m , according to examples of the present disclosure.

(8) FIG. 8 is an illustration of an apparatus for estimating a clock period or frequency by evaluating the first m samples and last m samples of a set of N time measurements, according to examples of the present disclosure.

(9) FIG. 9 is a more detailed illustration of an apparatus for estimating a clock cycle or period by evaluating the first m samples and last m samples of a set of N time measurements using two accumulators, according to examples of the present disclosure.

(10) FIG. 10 is an illustration of another possible implementation of an apparatus for estimating a clock cycle or period by evaluating the first m samples and last m samples of a set of N time measurements using a single accumulator, according to examples of the present disclosure.

(11) FIG. 11 is an illustration of yet another possible implementation of an apparatus for estimating a clock cycle or period by skipping L measurements after each sample, according to examples of the present disclosure.

(12) FIG. 12 is an illustration of a method for estimating a clock signal according to a least squares method, according to examples of the present disclosure.

(13) FIG. 13 is a more detailed illustration of a method for estimating a clock signal according to a least squares method, according to examples of the present disclosure.

(14) FIG. 14 is an illustration of a method for estimating a clock signal using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(15) FIG. 15 is a more detailed illustration of a method for estimating a clock signal using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(16) FIG. 16 is yet another more detailed illustration of a method for estimating a clock signal using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(17) FIG. 17 is an illustration of a method for estimating a clock signal by evaluating the first m samples and last m samples of a set of N time measurements, according to examples of the present disclosure.

(18) FIG. 18 is a more detailed illustration of a method for estimating a clock signal by evaluating the first m samples and last m samples of a set of N time measurements, according to examples of the present disclosure.

(19) FIGS. 19A and 19B an illustration of a method for estimating a clock signal by evaluating the first m samples and last m samples of a set of N time measurements and skipping L measurements, according to examples of the present disclosure.

DETAILED DESCRIPTION

(20) FIG. 2 is an illustration of an apparatus **200** for estimating a clock cycle or period according to a least squares method, according to examples of the present disclosure.

(21) A clock period may be estimated based on making time measurements of individual clock

periods or cycles over a predetermined time or for a predetermined number of periods or cycles. The time measurement of a given clock period or cycle may be made at a respective time of a respective portion of a given cycle or period of a clock signal, such as the rising edge or the falling edge of a clock signal. In the examples of the present disclosure, a rising edge may be used as the example time of an example portion of a given cycle or period of a clock cycle at which to make a time measurement, however this is not meant to be limiting in any way, and the falling edge may be used without exceeding the scope of the present disclosure. Furthermore, in examples of the present disclosure, such a time measurement may be already made using any suitable mechanism (not shown). Upon each such rising edge, for example, a time measurement of the rising edge may be made. The measurement of such a time may be made with a frequency that is sufficiently fast compared to the expected range of frequencies of the clock signal to be estimated. Moreover, the examples of the present disclosure may operate with a frequency or internal clock (not shown) that is sufficiently fast compared to the expected range of frequencies of the clock signal to be estimated. The time measurement of a given rising edge of the clock signal to be estimated may be given as $t_{\text{sub}.k}$. If N clock periods are to be measured, and thus N time measurements of a clock signal are to be made in order to estimate the frequency or period of the clock signal, $t_{\text{sub}.k}$ may be given as $t_{\text{sub}.k} \in \{t_{\text{sub}.0} \dots t_{\text{sub}.N-1}\}$. Each $t_{\text{sub}.k}$ may thus represent the time at which a given measurement was made of the clock signal to be estimated.

(22) The estimation of the clock period as performed by apparatus **200** using a least squares method may be defined as

$$(23) \quad \hat{T} = \frac{12 \cdot \text{Math}_{k=0}^{N-1} kt_k - 6(N-1) \cdot \text{Math}_{k=0}^{N-1} t_k}{N(N^2 - 1)} \quad \text{equation3}$$

Equation 3 may be derived from the least-squares approximation method.

(24) Apparatus **200** may be implemented in any suitable manner and within any suitable larger components or contexts. For example, apparatus **200** may be implemented within a microchip, microcontroller, clock regulation circuit, a system on a chip, or within any other suitable mechanism.

(25) Apparatus **200** may include a clock signal input **206** on which time measurements may be received. To estimate the frequency or period of a clock signal **212** (see also FIG. 1), N time measurements may be received on clock signal input **206**. The N time measurements may be received one at a time as they are measured from clock signal **212**. Each time measurement, denoted by $t_{\text{sub}.k}$, may arise from a respective time of a respective portion of a given clock cycle of clock signal **212** to be measured. For example, each $t_{\text{sub}.k}$ may be generated upon a rising edge of clock signal **212**. Each $t_{\text{sub}.k}$ may thus arrive at apparatus **200** upon a rising edge of clock signal **212**. Each $t_{\text{sub}.k}$ may arrive on clock signal input **206** in a serial or sequential manner, after the rising edge of clock signal **212**.

(26) Apparatus **200** may include a sampling circuit **202** and an estimation circuit **204** to generate an estimated frequency or period of clock signal **212** based upon the time measurements received on clock signal input **206**. Sampling circuit **202** and estimation circuit **204** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Although the present disclosure describes certain components and functions of sampling circuit **202** and estimation circuit **204**, respectively, variations may be implemented where some components or functions of sampling circuit **202** or estimation circuit **204** may be included in or performed by the other of sampling circuit **202** and estimation circuit **204**.

(27) The received $t_{\text{sub}.k}$ may be received by a sampling circuit **202**, which may sample $t_{\text{sub}.k}$. In various examples, sampling circuit **202** may selectively sample $t_{\text{sub}.k}$, wherein certain samples may be skipped, as discussed further below. From the received time measurements included in the various received $t_{\text{sub}.k}$, and after N such time measurements have been received (that is, $t_{\text{sub}.0}$

through $t_{\text{sub}.N-1}$), sampling circuit **202** may output a first sampled window **208** and a second sampled window **210** to estimation circuit **204**.

(28) First sampled window **208** may include an accumulation of the N time measurements received on clock signal input **206** from N consecutive measurements made of clock signal **212** by sampling circuit **202**. The accumulation may be performed in any suitable manner, such as by an accumulator.

(29) Second sampled window **210** may include an accumulation of products. The products may each be a product of a respective factor and a respective one of the N time measurements received on clock signal input **206** from N consecutive measurements made of clock signal **212**. The respective factors may each be the same or different. At least one of the respective factors may be greater than one. For example, the respective factor may be an index of the given time measurement. Thus, the given product may be the value $(k \cdot t_{\text{sub}.k})$. The accumulation may be performed in any suitable manner, such as by an accumulator.

(30) Sampled windows **208**, **210** may correspond to the generation of components of equation 3. For example, first sampled window **208** may be the term $\sum_{k=0}^{N-1} t_{\text{sub}.k}$ and second sampled window **210** may be the term $\sum_{k=0}^{N-1} k \cdot t_{\text{sub}.k}$. The factors 12 or 6 to these terms as shown in equation 3 may be applied by sampling circuit **202** or may be applied by estimation circuit **204**.

(31) Estimation circuit **204** may estimate the frequency or period of clock signal **212** from sampled windows **208**, **210**. Estimation circuit **204** may make such an estimate in any suitable manner, such as with a least squares estimation. Estimation circuit **204** may make such an estimate according to, for example, equation 2. Estimation circuit **204** may thus produce a value T for the estimated frequency of clock signal **212**, or a frequency, given by $1/T$ for the estimated frequency of clock signal **212**.

(32) FIG. 3 is a more detailed illustration of an apparatus **300** for estimating a clock cycle or period according to a least squares method, according to examples of the present disclosure. Apparatus **300** may be a more detailed implementation of apparatus **200**. Apparatus **300** may include a clock signal input **306** corresponding to clock signal input **206**, a sampling circuit **302** corresponding to sampling circuit **202**, a first sampled window **308** corresponding to first sampled window **208**, a second sampled window **310** corresponding to second sampled window **210**, and may estimate a frequency or a period of clock signal **312**, clock signal **312** corresponding to clock signal **212**.

(33) Apparatus **300** may include control circuit **322**. Control circuit **322** may orchestrate or control various aspects of apparatus **300** to provide the correct order of operations of the components so as to estimate the frequency or period of clock signal **312**. Control circuit **322** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof.

(34) Sampling circuit **302** of apparatus **300** may include a counter **334**. Counter **334** may be implemented in any suitable manner, such as a count-up counter, count-down counter, or timer. Counter **334** may count a quantity of time measurements $t_{\text{sub}.k}$ that have arrived on clock signal input **306** to make a given estimation of frequency or period of clock signal **312**. Counter **334** may thus increment an index k of the quantity of the N time measurements. For each clock cycle of clock signal **312**, counter **334** may increment k as the index of received time measurements. The value of k may be used as the respective factor to generate first sampled window **308** and second sampled window **310**. Counter **334** may be reset by control circuit **322** upon making a new estimation for clock signal **312**. Counter **334** may count the cycles of clock signal **312**. The output of counter **334** may be controlled by control circuit **322** so as to be an integer value between zero and $N-1$, wherein N is the total number of input clock cycles for which the frequency or period of clock signal **312** is to be estimated.

(35) Apparatus **300** may include multipliers **314**, **326**, **328**, and adder **330**. Each of these

mechanisms may be implemented in any suitable manner. Multiplier **314** may be part of sampling circuit **302**, and multipliers **326**, **238** and adder **300** may be part of estimation circuit **304**.

(36) Sampling circuit **302** of apparatus **300** may include a first accumulator **318** and a second accumulator **320**. Accumulators **318**, **320** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Specifically, accumulators **318**, **320** may be implemented by an adder accepting one input to be accumulated, and one input from a delay circuit (denoted as the block labeled $Z\{\text{circumflex over ()}\}-1$) routed in feedback from the output of the respective adder. The delay circuit may include a register or other storage for holding the running total of previous accumulated inputs.

(37) Clock signal input **306** may be multiplied with the currently produced index of counter **334** by multiplier **314**. This may be performed in sampling circuit **302** each time a new clock signal input **306** arrives. Thus, the given respective factor used to generate a given product to be accumulated by second accumulator **320** may be k . This product may be provided to second accumulator **320** to accumulate the product $k \cdot t_{\text{sub}.k}$. Thus, a running total of products may be accumulated in second accumulator **320** to generate second sampled window **310** as its output.

(38) Clock signal input **306** may be provided in parallel to first accumulator **318**. First accumulator **318** may accumulate a given time measurement $t_{\text{sub}.k}$ as it is received on clock signal input **306**. Thus, the given time measurement $t_{\text{sub}.k}$, as it is received on clock signal input **306** may be accumulated into a running total of the N time measurements to generate first sampled window **308**.

(39) Sampling circuit **302** of apparatus **300** may include a comparator **316**. Comparator **316** may be implemented in any suitable manner. Comparator **316** may compare the output of counter **334** with the quantity $N-1$ to determine whether N time measurements have been received.

(40) Sampling circuit **302** of apparatus **300** may include a switch **324**. Switch **324** may be implemented in any suitable manner, such as by a parallel set of transistors, switch fabric or switch matrix, or multiplexers. Switch **324** may receive a determination from comparator **316** that N time measurements have been received. Switch **324** may isolate the output of accumulators **318**, **320** until N time measurements have been received. When N time measurements have been received, switch **324** may connect the output of accumulators **318**, **320** to estimation circuit **304**. The output of accumulators **318**, **320** may manifest first sampled window **308** and second sampled window **310**, respectively.

(41) First sampled window **308** and second sampled window **310** may each be multiplied by a respective least squares factor according to equation 3. For example, first sampled window **308** may be multiplied by 12 by multiplier **12**. Second sampled window **310** may be multiplied by a factor of 6 ($N-1$) by multiplier **328**. This multiplication may be done during or after the generation of first sampled window **308** and second sampled window **310**. Thus, estimation circuit **304** may receive a scaled version of first sampled window **308** and a scaled version of second sampled window **310**. The results may be provided to adder **330**, in which the output of multiplier **326** is provided to a positive input of adder **330** and the output of multiplier **328** is provided to a negative input of adder **330**, and adder **330** may produce the difference. Thus, estimation circuit **304** may add scaled versions of sampled windows **308**, **310** to yield a sum or difference. The result output of adder **330** may be provided to multiplier **332**, which may apply N to the output of adder **330** to estimate the frequency or period of clock signal **312**. For example, multiplier **332** may multiply the output of adder **330** by a factor $1/(N\{\text{circumflex over ()}\}^2-1)$. The result may be the estimated period of clock signal **312**, and the inverse of which may be the estimated frequency of clock signal **312**.

(42) The estimation error for the least square method of FIGS. 2-3 may be given as

(43)
$$\sigma^2 = \frac{12}{N(N^2 - 1)}$$
 equation 4

where each rising edge of the clock signal is assumed to have independent measurement noise with variance of σ^2 . The estimation error thus decreases with increasing measurement window T (and increase of N) and the error reduction is proportional to $1/N^3$, faster than the method of equation 1 (that is, $1/N^2$). Thus, techniques and apparatuses of FIGS. 2-3 may represent a more accurate manner of estimating the frequency or period of a clock signal.

(44) As shown and discussed above, apparatus **300** may include multiplication by multiplier **314** of each received time measurement $t_{sub.k}$ before accumulating the resulting product. This amount of multiplication, wherein each time measurement $t_{sub.k}$ is to be multiplied by a factor k before being accumulated, may cause an increase in the computational complexity of estimating the period or frequency of clock signal **312**.

(45) FIG. 4 illustrates an apparatus **400** for estimating a clock period or frequency using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure. In comparison to the apparatuses of FIGS. 2-3, apparatus **400** may provide more efficient estimation by reducing a need to multiply each incoming time measurement $t_{sub.k}$ as it arrives before accumulating it as part of a least squares estimation.

(46) The estimation of the period or frequency of the clock signal of apparatus **400** may be given as

(47)
$$\hat{T} = \frac{1}{N-m} \cdot \sum_{k=m}^{N-1} (t_k - t_{k-m})$$
 equation 5

where m is a delay for a given time measurement.

(48) Apparatus **400** may be implemented in any suitable manner and within any suitable larger components or contexts. For example, apparatus **400** may be implemented within a microchip, microcontroller, clock regulation circuit, a system on a chip, or within any other suitable mechanism. Apparatus **400** may be implemented in a same or different mechanism as apparatuses **200**, **300**. Apparatus **400** may receive a clock signal input **406**, which may be an example of clock signal inputs **206** or **306**. Time measurements may be received. To estimate the frequency or period of a clock signal **412**, N time measurements may be received on clock signal input **406**. The N time measurements may be received one at a time as they are measured from clock signal **412**. Each time measurement, denoted by $t_{sub.k}$, may arise from a respective time of a respective portion of a given clock cycle of clock signal **412** to be measured. For example, each $t_{sub.k}$ may be generated upon a rising edge of clock signal **412**. Each $t_{sub.k}$ may thus arrive at apparatus **400** upon a rising edge of clock signal **412**. Each $t_{sub.k}$ may arrive on clock signal input **406** in a serial or sequential manner, after the rising edge of clock signal **412**.

(49) Apparatus **400** may include a sampling circuit **402** and an estimation circuit **404** to generate an estimated frequency or period of clock signal **412** based upon the time measurements received on clock signal input **406**. Sampling circuit **402** and estimation circuit **404** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Although the present disclosure describes certain components and functions of sampling circuit **402** and estimation circuit **404**, respectively, variations may be implemented where some components or functions of sampling circuit **402** or estimation circuit **404** may be included in or performed by the other of sampling circuit **402** and estimation circuit **404**.

(50) The received $t_{sub.k}$ may be received by sampling circuit **402**, which may sample time measurement $t_{sub.k}$. In various examples, sampling circuit **402** may selectively sample time measurement $t_{sub.k}$, wherein certain samples may be skipped, as discussed further below. For each received time measurement $t_{sub.k}$ above a delay value m , sampling circuit **402** may output a sampled window to estimation circuit **404**. Such a set of sampled windows may include at least a first sampled window **408** and a second sampled window **410**. Each sampled window may be given as the difference between the time measurements $t_{sub.k}$ and $t_{sub.k-m}$, where again, m is a delay

to reference a prior time measurement. The quantity of sampled windows may thus be a quantity of up to $(N-m)$. Sampled windows may be sent one at a time each clock cycle of clock signal **412**. In some examples, sampled windows may be sent only after k reaches m . In other examples, sampled windows may be sent before k reaches m , but such sampled windows might be discarded.

(51) Thus, upon receiving a first time measurement $t_{\text{sub}.k}$ on clock signal input **406**, sampling circuit **402** may generate first sampled window **408**. First sampled window **408** may be based upon the first time measurement and a first previous time measurement. The previous time measurement may be delayed with respect to the first time measurement by m time measurements **406** or samples of clock signal input **406** or clock cycles of clock signal **412**. The quantity m may thus also represent a count of delayed measurements m . For example, at a given time x , the given time measurement on **406** may be $t_{\text{sub}.x}$, wherein x is greater than m , and first sampled window **408** may be $(t_{\text{sub}.x} - t_{\text{sub}.x-m})$. Expressed differently, first sampled window **408** may be $(t_{\text{sub}.k} - t_{\text{sub}.k-m})$.

(52) Upon receiving a second time measurement subsequent to the first time measurement on clock signal input **406**, sampling circuit **402** may generate second sampled window **410**. Second sampled window **410** may be based upon the second time measurement and a second previous time measurement. The second previous time measurement may be delayed with respect to the second time measurement by the same number, m , which was used in generating the first sampled window. The quantity m may represent time measurements **406** or samples of clock signal input **406** or clock cycles of clock signal **412**. For example, at a given time y , the given time measurement on **406** may be $t_{\text{sub}.y}$, wherein y is greater than x , which is greater than m , and first sampled window **408** may be $(t_{\text{sub}.y} - t_{\text{sub}.y-m})$. Expressed differently, second sampled window **410** may be $(t_{\text{sub}.k+1} - t_{\text{sub}.k+1-m})$.

(53) The process of receiving time measurements on clock signal input **406** and generating sampled windows may continue until $(N-m)$ sampled windows are generated.

(54) Sampled windows **408**, **410** may correspond to the generation of components of equation 5. For example, each may represent the $(t_{\text{sub}.k} - t_{\text{sub}.k-m})$ that is to be summed in equation 5. The sampled windows **408**, **410** may be scaled the factor $1/m$, or such scaling may be performed by estimation circuit **404**.

(55) Estimation circuit **404** may estimate the frequency or period of clock signal **412** from sampled windows **408**, **410**, and upon any other suitable available sampled windows. Estimation circuit **404** may make such an estimate in any suitable manner, such as with, for example, equation 5. Estimation circuit **404** may thus produce a value T for the estimated frequency of clock signal **412**, or a frequency, given by $1/T$ for the estimated frequency of clock signal **412**.

(56) FIG. 5 is a more detailed illustration of an apparatus **500** for estimating a clock cycle or period using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure. Apparatus **500** may be a more detailed implementation of apparatus **400**. Apparatus **500** may include a clock signal input **506** corresponding to clock signal input **406**, a sampling circuit **502** corresponding to sampling circuit **402**, a sampled window **522** corresponding to first sampled window **408** and second sampled window **410**, and may estimate a clock signal **512** corresponding to clock signal **412**.

(57) Apparatus **500** may include delay circuits **520**, **514**, delay circuits **514** formed as a chain **524** of delay circuits, adders **508**, **516**, and multipliers **510**, **518**. Moreover, all or a portion of estimation circuit **504** may make up a finite impulse response (FIR) filter **521**. Each of these may be implemented in any suitable manner. Sampling circuit **502** may comprise delay circuit **520**, adder **508** and multiplier **510**. Estimation circuit **504** may comprise delay circuits **514**, adder **516** and multiplier **518**.

(58) For example, delay circuits **520**, **514** may be implemented as a register or other suitable memory location. Each of delay circuits **520**, **514** may store a previously stored value for a designated number of cycles of clock signal **512** or samples thereof. For example, delay circuit **520**

may be controlled by a designation of “-m” to store a value that was input m cycles previously. Thus, delay circuit **520**, at a given cycle k of clock signal **512**, may output the value it was loaded with at the cycle (k-m). Delay circuits **514** may be implemented with a designation of “-1” such that each returns, at a given cycle k of clock signal **512**, the value that was input at cycle (k-1).

(59) In sampling circuit **502**, time measurements $t_{\text{sub},k}$ on clock signal input **506** may be loaded into delay circuit **520** at a given cycle k. Delay circuit **520** may yield the value with which it was loaded in the mth previous cycle, given as $t_{\text{sub},k-m}$. The value $t_{\text{sub},k-m}$ may be a first previous time measurement for a first time measurement $t_{\text{sub},k}$. The value $t_{\text{sub},k-m+1}$ may be a second previous time measurement for a second time measurement $t_{\text{sub},k+1}$.

(60) Upon receiving time measurements $t_{\text{sub},k}$, time measurements $t_{\text{sub},k}$ may be stored in delay circuit **520** and delay circuit **520** may return the previous time measurement of $t_{\text{sub},k-m}$. Adder **508**, which may receive time measurement $t_{\text{sub},k}$ on its positive input and the output of delay circuit **520** on its negative input, may subtract $t_{\text{sub},k-m}$ from $t_{\text{sub},k}$. The result may be divided by m by multiplier **510**, i.e. the result is multiplied by a factor $1/m$. The result may be a sampled window **522**, such as first sampled window **408**.

(61) Upon receiving $t_{\text{sub},k+1}$, $t_{\text{sub},k+1}$ may be stored in delay circuit **520** and delay circuit **520** may return the previous time measurement of $t_{\text{sub},k+1-m}$. Adder **508** may subtract $t_{\text{sub},k+1-m}$ from $t_{\text{sub},k+1}$. The result may be divided by m by multiplier **510**. The result may be a sampled window **522**, such as second sampled window **410**.

(62) The results of sampling circuit **502** may be produced as sampled windows. As a given result of sampling circuit **502** is produced, it may be stored in a memory location such as a delay circuit of chain **524**. In one example, as a given result of sampling circuit **502** is produced, it may be stored in a first or head delay circuit **514A** of chain **524** of delay circuits of estimation circuit **504**. At each cycle of clock signal **512**, upon receiving a given sampled time measurement on clock signal input **506**, delay circuits **514** may propagate its value to a subsequent delay circuit or next delay circuit **514** in chain **524**. First or head delay circuit **514A** may receive its value as a sampled window **522** from sampling circuit **502**. The remainder of chain **524** may receive its value from the next highest delay circuit **514** in chain **524**. Thus, sampling circuit **502** is to store the first sampled window from $(t_{\text{sub},k}-t_{\text{sub},k-m})$ in delay circuit **514A** upon receiving a first time measurement $t_{\text{sub},k}$ on clock signal input **506**. Upon receiving a second time measurement $t_{\text{sub},k+1}$, delay circuit **514A** is to propagate the first sampled window to a next delay circuit **514B**, and sampling circuit **502** is to store the second sampled window from $(t_{\text{sub},k+1}-t_{\text{sub},k+1-m})$ in delay circuit **514A**.

(63) The process of sampling time measurements by sampling circuit **502** from clock signal input **506** may continue until N time measurements have been sampled. This may result in the generation and storage of a quantity (N-m) sampled windows in chain **524**.

(64) As discussed above, estimation circuit **504** may include or be implemented as FIR filter **521**, which may estimate, from at least the first and second sampled windows, the frequency or period of clock signal **512**. FIR filter **521** may include a quantity of (N-m) taps, which may be manifested by inputs to adder **516** from the (N-m) delay circuits **514** of chain **524**. Adder **516** may add the values stored in the (N-m) delay circuits **514** of chain **524**, and multiplier **518** may divide the result by (N-m), i.e. multiply the result of adder **516** by a factor $1/(N-m)$.

(65) Generally speaking, sampling circuit **502** thus generates a given sampled window **522** upon receiving a given time measurement $t_{\text{sub},k}$ on clock signal input **506**. Such generation may include first sampled window **408** and second sampled window **410** of FIG. 4. Sampling circuit **502** may store sampled window **522** in delay circuit **514A** of chain **524** of delay circuits. Sampling circuit **502** may perform such generation for at least a quantity (N-m) successive time measurements.

(66) Delay circuit **514A** may propagate an immediately previously generated sampled window to a next successive delay circuit **514B** in chain **524** of delay circuits upon reception of a given time measurement $t_{\text{sub},k}$ on clock signal input **506**. The other delay circuits **514** in chain **524** may also propagate their immediately previously stored sampled window to a next successive delay circuit

514 in chain 524.

(67) After reception of N successive time measurements (to . . . t.sub.N-1) on clock signal input 506, estimation circuit may use (N-m) sampled windows as stored in (N-m) delay circuits 514 in chain 524 to estimate the frequency or period of clock signal 512.

(68) FIG. 6 is another illustration of operation of apparatus 500 using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure.

(69) The operation of sampling circuit 502 to generate a given sampled window 522 may also produce an estimation of the period or frequency for a portion of the sample time Ts for the N time measurements. The portion of the sample time Ts for the N time measurements may be denoted as Ts', and may be referred to as a partial time period. The partial time period Ts' may include the time between a given time measurement t.sub.k and a delayed time measurement t.sub.k-m delayed by m measurements with respect to the given time measurement t.sub.k. Sampling circuit 502, by subtracting the delayed time measurement t.sub.k-m from the given time measurement t.sub.k, and dividing it by m, may be an estimate of the clock period of clock signal 512 over the partial time period Ts'. This estimate may be performed according to equation 1. Thus, a given sampled window 522 may be an estimate of the clock period of clock signal 512 over the given partial time period, which is the time between the given time measurement and the delayed time measurement.

(70) As discussed above, sampling circuit 502 may store sampled window 522 in chain 524 of delay circuits 514.

(71) Moreover, estimation circuit 504 may, upon reception of a quantity N successive time measurements at clock signal input 506, average the quantity (N-m) stored estimations of the frequency or period of clock signal 512 for the given partial time periods, as stored in delay circuits 514.

(72) FIG. 7 is an illustration of simulation of error response for various values of delay m, according to examples of the present disclosure.

(73) With regards to FIGS. 4-6, the selection of a value for m may affect the estimation error.

(74) The estimation error when m is greater than or equal to N/2 may be given as

(75)
$$\sigma^2 = \frac{2}{m^2(N-m)} \quad \text{equation 6}$$

(76) For m less than N/2, there may be overlap between t.sub.k+m and t.sub.k, which will cancel each other for k=m to k=N-m-1. Hence, the estimation error may be given as

(77)
$$\sigma^2 = \frac{2}{m(N-m)^2} \quad \text{equation 7}$$

Equations 6 and 7 may be obtained from static analysis of error distribution with an assumption of independent and identical distributed observation noise.

(78) Thus, as shown in FIG. 7, it can be observed that the estimation error does not just depend on the number of collected clock rising edges (N) but also the choice of m. FIG. 7 shows the relationship between clock period estimation error and values of m when N=1000. The estimation error for the least squares technique of FIGS. 2-3 is also shown for comparison. It can be seen from FIG. 7 that the performance of the apparatuses of FIGS. 4-6 can be very close to that of the least square's method of FIGS. 2-3 with a correct choice of m.

(79) To get an improved value of m, the differential of ϵ^2 can be taken with respect to m and then set it to 0:

(80)
$$\frac{\partial}{\partial m} \sigma^2 = 0.$$

As a result, for $m \geq N/2$, a good value for m may be $m=2N/3$ and the estimation error may be

(81)
$$\sigma^2 = \frac{27}{2N^3} \quad \text{equation 8}$$

(82) For $m < N/2$, the good value for m may be $m=N/3$ and the estimation error may be

(83)
$$\sigma^2 = \frac{27}{2N^3} \quad \text{equation 9}$$

(84) For both scenarios, the estimation error may be the same and may be proportional to $1/N\{\text{circumflex over ()}\}^3$.

(85) FIG. 8 is an illustration of an apparatus 800 for estimating a clock period or frequency by evaluating the first m samples and last m samples of a set of N time measurements, according to examples of the present disclosure. In comparison to the apparatuses of FIGS. 4-7, apparatus 400 may provide more efficient estimation by reducing a need to include a quantity of (N-m) taps for the FIR filter and estimation circuit therein.

(86) The estimation of the period or frequency of the clock signal of apparatus 800 may begin with a restatement again of equation 5, and rewriting it as follows to yield equation 10:

(87) $\hat{T} = \frac{1}{N-m} \cdot \text{Math.}_{k=m}^{N-1} \frac{1}{m}(t_k - t_{k-m})$ equation 5 rewritten as

$$\hat{T} = \frac{1}{m(N-m)} \left(\text{Math.}_{k=m}^{N-1} t_k - \text{Math.}_{k=m}^{N-1} t_{k-m} \right) \text{rewritten as}$$

$$\hat{T} = \frac{1}{m(N-m)} \left(\text{Math.}_{k=m}^{N-1} t_k - \text{Math.}_{k=0}^{N-m-1} t_k \right) \text{ equation 10}$$

where m is a change in sample size.

(88) Apparatus 800 may be implemented with two accumulation processes, each with different start and end points. But the two accumulation processes share the same elements from {m . . . N-m-1} if $m < (N-1)/2$, and these same elements are cancelled out after subtraction. Thus, equation 10 can be further written as

(89) $\hat{T} = \frac{1}{m(N-m)} \left(\text{Math.}_{k=N-m}^{N-1} t_k - \text{Math.}_{k=0}^{m-1} t_k \right)$ equation 11

(90) Thus, two accumulation processes are still used. The estimation of the frequency or period of clock signal 812 may be proportional to the difference between the sum of the last m time measurements and the sum of first m time measurements.

(91) Apparatus 800 may include an architecture to perform these processes. In various further examples, shown further below, the two accumulation processes might be performed with one or two accumulators.

(92) Apparatus 800 may be implemented in any suitable manner and within any suitable larger components or contexts. For example, apparatus 800 may be implemented within a microchip, microcontroller, clock regulation circuit, a system on a chip, or within any other suitable mechanism. Apparatus 800 may be implemented in a same or different mechanism as apparatuses 200, 300, 400, 500. Apparatus 800 may receive a clock signal input 806, which may be an example of clock signal inputs 206, 306, 406, 506. Time measurements $t_{\text{sub}.k}$ may be received. To estimate the frequency or period of a clock signal 812, N time measurements $t_{\text{sub}.k}$ may be received on clock signal input 806. The N time measurements $t_{\text{sub}.k}$ may be received one at a time as they are measured from clock signal 812. Each time measurement, denoted by $t_{\text{sub}.k}$, may arise from a respective time of a respective portion of a given clock cycle of clock signal 812 to be measured. For example, each $t_{\text{sub}.k}$ may be generated upon a rising edge of clock signal 812. Each $t_{\text{sub}.k}$ may thus arrive at apparatus 800 upon a rising edge of clock signal 812. Each $t_{\text{sub}.k}$ may arrive on clock signal input 806 in a serial or sequential manner, after the rising edge of clock signal 812.

(93) Apparatus 800 may include a sampling circuit 802 and an estimation circuit 804 to generate an estimated frequency or period of clock signal 812 based upon the time measurements received on clock signal input 806. Sampling circuit 802 and estimation circuit 804 may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Although the present disclosure describes certain components and functions of sampling circuit 802 and estimation circuit 804, respectively, variations may be implemented where some components or functions of sampling circuit 802 or estimation circuit 804 may be included in or performed by the other of sampling circuit 802 and estimation circuit 804.

(94) The received time measurement $t_{\text{sub},k}$ may be received by sampling circuit **802**, which may sample $t_{\text{sub},k}$. In various examples, sampling circuit **802** may selectively sample $t_{\text{sub},k}$, wherein certain samples may be skipped, as discussed further below.

(95) Sampling circuit **802** may generate a first sampled window **808**. First sampled window **808** may include a sum of two or more of the first m elements of the N time measurements received on clock signal input **806**. In one example, all of the first m elements of the N time measurements received on clock signal **806** may be summed and included in first sampled window **808**. In another example, some of these time measurements may be skipped. The set of the first m time measurements of the N time measurements that are summed may be denoted $t_{\text{sub},0} \dots t_{\text{sub},m-1}$, and the sum represented in first sampled window **808** may be denoted as $\sum_{k=0}^{m-1} t_{\text{sub},k}$. This may be the accumulated first m time measurements of the N time measurements that arrive on clock signal input **806**.

(96) Sampling circuit **802** may generate a second sampled window **810**. Second sampled window **810** may include a sum of two or more of the last m of the N time measurements received on clock signal input **806**. In one example, all of the elements of the last m of the N time measurements $t_{\text{sub},k}$ received on clock signal **806** may be summed and included in second sampled window **810**. In another example, some of these time measurements may be skipped. The set of the last m elements of the time measurements of the N time measurements may be denoted $t_{\text{sub},N-m} \dots t_{\text{sub},N-1}$. The quantity of these time measurements may be m . The sum represented in second sampled window **810** may be denoted as $\sum_{k=N-m}^{N-1} t_{\text{sub},k}$. This may be the accumulated last m time measurements of the N time measurements that arrive on clock signal input **806**.

(97) Estimation circuit **804** may estimate the frequency or period of clock signal **812** based upon first sampled window **808** and second sampled window **810**. Estimation circuit **804** may make such an estimation using, for example, equation 11. Estimation circuit **804** may subtract sampled window **808** from sampled window **810**, and apply a factor of $1/(m(N-m))$ to the result to obtain the estimation of the frequency or period of clock signal **812**,

(98) FIG. 9 is a more detailed illustration of an apparatus **900** for estimating a clock cycle or period by evaluating the first m samples and last m samples of a set of N time measurements using two accumulators, according to examples of the present disclosure. Apparatus **900** may be a more detailed implementation of apparatus **800**. Apparatus **900** may include a clock signal input **906** corresponding to clock signal input **806**, a sampling circuit **902** corresponding to sampling circuit **802**, a first sampled window **908** corresponding to first sampled window **808**, a second sampled window **910** corresponding to second sampled window **810**, and may estimate a clock signal **912** corresponding to clock signal **812**.

(99) Apparatus **900** may include switches **914**, **916**, **918**, a first accumulator **924**, a second accumulator **926**, counter **920**, control circuit **922**, adder **928**, and multiplier **930**. Each of these may be implemented in any suitable manner. Sampling circuit **902** comprises switches **914**, **916**, **918**, first accumulator **924**, second accumulator **926**, counter **920** and control circuit **922**. Estimation circuit **904** comprises adder **928** and multiplier **930**.

(100) Switches **914**, **916**, **918** may be implemented in any suitable manner, such as by a parallel set of transistors, switch fabric or switch matrix, or multiplexers.

(101) Accumulators **924**, **926** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Specifically, accumulators **924**, **926** may be implemented by an adder accepting one input to be accumulated, and one input from a delay circuit (denoted as the block labeled $Z\{\text{circumflex over ()}\}-1$) routed in feedback from the output of the adder. The delay circuit may include a register or other storage for holding the running total of previous accumulated inputs.

(102) Counter **920** may be implemented in any suitable manner, such as a count-up counter, count-

down counter, or timer. Counter **920** may count a quantity of time measurements $t_{\text{sub}.k}$ that have arrived on clock signal input **906** to make a given estimation of frequency or period of clock signal **912**. Counter **920** may thus increment an index k of the quantity of the N time measurements. Each clock cycle of clock signal **912**, counter **920** may increment k as the index of received time measurements. Counter **920** may count the cycles of clock signal **912**. The output of counter **920** may be controlled so as to be an integer value between zero and $N-1$, wherein N is the total number of input clock cycles for which the frequency or period of clock signal **912** is to be estimated.

(103) Control circuit **922** may orchestrate or control various aspects of apparatus **900** to provide the correct order of operations of the components so as to estimate the frequency or period of clock signal **912**. Control circuit **922** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof.

(104) Control circuit **922** may reset counter **920** and accumulators **924**, **926** upon starting a new determination of a frequency or period of clock signal **912**.

(105) Time measurements may arrive on clock signal input **906**, denoted as $t_{\text{sub}.k}$.

(106) For the first m time measurements of the N time measurements that will arrive for the estimation of a frequency or period of clock signal **912**, as indicated by counter **920**, control circuit **922** may close switch **914** to allow time measurements to be applied to an input of first accumulator **924** so as to accumulate these first m values. These values accumulated in accumulator **924** may be a running total of the first m received time measurements to be produced as first sampled window **908**. Initially, control circuit **922** may open switch **916** so as to prevent time measurements to be applied to an input of second accumulator **926** until the last m values in the N time measurements arrive. Also, switch **918** may be opened by control circuit **922** so as to prevent output of first and second accumulators **924**, **926** from being provided as sampled windows **908**, **910** to estimation circuit **904**. When m time measurements have been accumulated, control circuit **922** may open switch **914** to prevent further time measurements to be accumulated in first accumulator **924**.

(107) Thus, upon reception of a given time measurement $t_{\text{sub}.k}$ on clock signal input **906**, if less than m measurements have been received, first accumulator **924** will add the given time measurement to a first running total of time measurements, which may be eventually produced as sampled window **908**. Upon reception of a given time measurement $t_{\text{sub}.k}$ on clock signal input **906**, if m or greater time measurements have been received, first accumulator **924** may maintain its first running total of time measurements. First accumulator **924** may thus add successive time measurements of the first m of the N time measurements to generate sample window **908**.

(108) For the last m of the N time measurements that will arrive for the estimation of a frequency or period of clock signal **912**, as indicated by counter **920**, control circuit **922** may close switch **916** so as to allow time measurements to be applied to an input of second accumulator **926** so as to accumulate these last m of the N time measurements therein. These values accumulated in second accumulator **926** may be a running total of the last m received time measurements to be produced as second sampled window **910**. Also, switch **918** may continue to be open so as to prevent output of first and second accumulators **924**, **926** from being provided as sampled windows **908**, **910** to estimation circuit **904**. The operation of control circuit **922** to open or close switch **914** may be independent, as it may be performed according to whether the first m time measurements have arrived.

(109) Thus, upon reception of a given time measurement $t_{\text{sub}.k}$ on clock signal input **906**, if $(N-m)$ or greater time measurements and less than N time measurements have been received, second accumulator **926** may add the given time measurement to a second running total of time measurements to generate second sampled window **910**.

(110) Once N time measurements have been received, control circuit **922** may open switch **918** so as to allow the values accumulated in accumulators **924**, **926** to be provided to estimation circuit **904**. These values may be denoted as first sampled window **908** and second sampled window **910**, respectively.

(111) In estimation circuit **904**, adder **928** may subtract first sampled window **908** from second sampled window **910**. In particular, when switch **918** is closed, the output of first accumulator **924** is provided to the negative input of adder **928** and the output of second accumulator **926** is provided to the positive input of adder **928**. The result may be divided by $(m(N-m))$ by multiplier **930**, i.e. the output of adder **928** may be multiplied by a factor $1/(m(N-m))$ by multiplier **930**, and the output given as the estimated period or frequency of clock signal **912**. Upon the sampling circuit reception of N time measurements, estimation circuit **904** may thus apply first sampled window **908** and second sampled window **910** to adder **928**, subtracting first sampled window **908** from second sampled window **910**, and apply a result from adder **928** to a multiplication factor of $1/(m(N-m))$ with multiplier **930** to yield an estimated a frequency or period of the clock signal.

(112) FIG. **10** is an illustration of another possible implementation of apparatus **900** for estimating a clock cycle or period by evaluating the first m samples and last m samples of a set of N time measurements using a single accumulator, according to examples of the present disclosure.

(113) Apparatus **900** as shown in FIG. **10** may differ in comparison with FIG. **9** in terms of the number of accumulators used. In FIG. **9**, if m is less than $(N-1)/2$, then switches **914**, **916** might not be ON at the same time and thus not accumulate a given same time measurement $t_{\text{sub}.k}$. Thus, a single accumulator might be used as shown in FIG. **10** when m is less than $N/2$.

(114) Apparatus **900** may still include a counter **920**, control circuit **922**, adder **928**, and multiplier **930**.

(115) Apparatus **900** may include switches **936**, **938**, which may be implemented in any suitable manner, such as by a parallel set of transistors, switch fabric or switch matrix, or multiplexers. Sampling circuit **902** may comprise switches **936**, **938**, accumulator **940**, counter **920** and control circuit **922**. Estimation circuit **904** may comprise adder **928** and multiplier **930**. Register **932** and **934** may be shared between sampling circuit **902** and estimation circuit **904**.

(116) Apparatus **900** may include a single accumulator **940**, which may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Specifically, accumulators **924**, **926** may be implemented by an adder accepting one input to be accumulated, and one input from a delay circuit (denoted as the block labeled $Z\{\text{circumflex over } (\)\}-1$) routed in feedback from the output of the adder. The delay circuit may include a register or other storage for holding the running total of previous accumulated inputs.

(117) Apparatus **900** may include memory locations such as a first register **932** and a second register **934**.

(118) Control circuit **922** may reset accumulator **940** when the estimation of the period or frequency of clock signal **912** begins. Moreover, control circuit **922** may reset accumulator **940** when m time measurements have been made from clock signal input **906**. Thus, control circuit **922** may reset accumulator **940** when counter **920** is at zero and at $(N-m)$.

(119) Accumulator **940** may selectively receive time measurements $t_{\text{sub}.k}$ from clock signal input **906**. The selective application of time measurements from clock signal input **906** may be made by operation of switch **936**. Moreover, output values of accumulator **940** may be selectively applied to one of registers **932**, **934** through operation of switch **938**. Switches **936**, **938** may be controlled by control circuit **922** based on the count of received time measurements as counted by counter **920**.

(120) When the estimation begins and the index produced by counter **920** is zero, switch **936** may be closed and time measurements $t_{\text{sub}.k}$ from clock signal input **906** may be applied to the input of accumulator **940**. Accumulator **940** may add received time measurements as a running total of a

sum of the first m time measurements. This may continue until counter **920** reaches $m-1$. At that time, switch **936** may be opened by control circuit **922** and switch **938** may be caused to connect the output of accumulator **940** to first register **932** so that the value of accumulator **940** may be stored in first register **932**. This may be first sampled window **908** produced by accumulator **940** and stored in first register **932**, to be provided to estimation circuit **904**.

(121) Accordingly, upon reception of a given time measurement $t_{sub.k}$ on clock signal input **906**, if counter **920** indicates that less than m time measurements have been received, accumulator **940** may add the given time measurement to the first running total of time measurements accumulated therein to generate first sampled window **908**. Upon reception of a given time measurement $t_{sub.k}$ on clock signal input **906**, if counter **920** indicates that m time measurements have been received, sampling circuit **902** is to store the first running total of time measurements as first sampled window **908** in a first memory such as first register **932** and to reset accumulator **940**.

(122) Subsequently, switch **936** may be open until the number of time measurements as counted by counter **920** reaches $(N-m)$. Then, accumulator **940** may be reset by control circuit **922**. Then, switch **936** may again be closed by control circuit **922** so as to again apply received time measurements $t_{sub.k}$ on clock signal input **906** to accumulator **940**, to accumulate the last m received time measurements of the N time measurements. Accumulator **940** may accumulate these values as a running total of the sum of the last m received time measurements, to be used to produce sampled window **910**.

(123) Accordingly, upon reception of a given time measurement $t_{sub.k}$ on clock signal input **906**, if counter **920** indicates that a quantity of time measurements greater than or equal to $(N-m)$ but less than N has been received, accumulator **940** may add the given time measurement to the second running total of time measurements accumulated therein to generate second sampled window **910**. Upon reception of a given time measurement $t_{sub.k}$ on clock signal input **906**, if counter **920** indicates that N time measurements have been received, sampling circuit **902** is to store the first running total of time measurements as second sampled window **910** in a second memory such as second register **934** and to reset accumulator **940**, by closing the appropriate portion of switch **938**, responsive to control circuit **922**.

(124) Upon the sampling circuit reception of N time measurements, estimation circuit **904** may apply first sampled window **908** as stored in first register **932** and second sampled window **910** as stored in second register **934** to adder **928**, subtracting first sampled window **908** from second sampled window **910**, and apply a result from adder **928** to a multiplication factor of $1/(m(N-m))$ with multiplier **930** to yield an estimated a frequency or period of the clock signal. In particular, the output of first register **932** is provided to the negative input of adder **928** and the output of second register **934** is provided to the positive input of adder **928**. The result may be divided by $(m(N-m))$ by multiplier **930**, i.e. the output of adder **928** may be multiplied by a factor $1/(m(N-m))$ by multiplier **930**,

(125) FIG. **11** is an illustration of yet another possible implementation of apparatus **900** for estimating a clock cycle or period by skipping L measurements after each sample, according to examples of the present disclosure. FIG. **11** illustrates a possible implementation or variation of any of the apparatuses of FIGS. **8-10**, wherein the sampling rate may be reduced by skipping the sampling of some of the received time measurements on clock signal input **906**. The illustration in FIG. **11** is specifically made with reference to a variation of the illustration in FIG. **10**, but may also be applicable to the illustrations of FIGS. **8-9**.

(126) In FIG. **11**, a skip circuit **950** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, programmable logic, instructions for execution by a processor, an application specific integrated circuit, a field programmable gate array, or any suitable combination thereof. Skip circuit **950** may selectively skip any suitable number of samples incoming on clock signal input **906**. A given time measurement $t_{sub.k}$ may be followed by a designated number of skipped time measurements. The number of time measurements to omit or skip may be given as L,

resulting in sampling of every (L+1)th time measurement. Thus, a given time measurement that is actually sampled for use in accumulation and estimation of frequency or period of clock signal **912** may be given as $t_{\text{sub.k}L}$. If L was 3, for example, the actually sampled time measurements would be $t_{\text{sub.k}}$, $t_{\text{sub.k}+4}$, $t_{\text{sub.k}+8}$, etc., in the form of $t_{\text{sub.k}+i(L+1)}$.

(127) The estimation of a period of clock signal **912** using this technique may be given as

$$(128) \quad \hat{T} = \frac{1}{\frac{N-m}{L} + 1} \cdot \text{Math.}_{k=\frac{m}{L}}^{\frac{N-1}{L}} \frac{1}{mL} (t_{kL} - t_{kL-m}) \quad \text{equation12}$$

which may be further reduced as

$$(129) \quad \hat{T} = \frac{1}{m(N-m+L)} \left(\text{Math.}_{k=\frac{N-m-1}{L}+1}^{\frac{N-1}{L}} t_{kL} - \text{Math.}_{k=0}^{\frac{m}{L}-1} t_{kL} \right) \quad \text{equation13}$$

when $m < N/2$, and a single accumulator can be used such as the implementation in FIG. **10**. By skipping a next L time measurements to sample, accumulator **940** may be operated at a lower clock rate.

(130) In FIG. **10**, sampling circuit **902** may thus use skip circuit **950** to skip L samples after a given received time measurement $t_{\text{sub.k}}$ before sampling another received time measurement, and to exclude the skipped L samples from generation of first sampled window **908** and second sampled window **910**. These skipped L samples may be excluded by being not sampled and not provided to accumulator **940** for accumulation in either the first or second running totals of the first or last m samples of the received time measurements.

(131) The set of received time measurements $t_{\text{sub.k}}$ may be of a size N, while the set of actually sampled received time measurements $t_{\text{sub.k}L}$ may be of a size N/L. Thus, sampling circuit **902** may sample a quantity N/L of the received N time measurements.

(132) Control circuit **922** may cause accumulator **940** to accumulate a first quantity of sampled time measurements $t_{\text{sub.k}L}$, such as (m/L), and store these in a first memory such as first register **932** as first sampled window **908**. Control circuit **922** may cause accumulator **940** to accumulate a last quantity (m/L) of sampled time measurements $t_{\text{sub.k}L}$, and store these in a second memory such as second register **934** as second sampled window **910**.

(133) Moreover, counter **920** operation may be adjusted to account for the skipping of L samples by skip circuit **950**. Thus, counter **920** may count a number of time measurements that are actually sampled—i.e., $t_{\text{sub.k}L}$ time measurements, as opposed to $t_{\text{sub.k}}$ time measurements. Moreover, this may cause the control of switches **936**, **938** and accumulator **940** to factor in L, the number of samples skipped between $t_{\text{sub.k}L}$ time measurements. Generally, the thresholds described in FIG. **10** may be augmented by dividing by L or L-1, as appropriate.

(134) Upon sampling a given time measurement $t_{\text{sub.k}L}$ from clock signal input **906** and forwarded by skip circuit **950**, if counter **920** indicates that less than m/L time measurements have been sampled, sampling circuit **902** may cause control circuit **922** to cause accumulator **940** to add the sampled given time measurement $t_{\text{sub.k}L}$ to a first running total of sampled time measurements to generate first sampled window **908**.

(135) Upon sampling a given time measurement $t_{\text{sub.k}L}$ from clock signal input **906** and forwarded by skip circuit **950**, if counter **920** indicates that more than m/L but less than (N-m)/L time measurements have been sampled, sampling circuit **902** may cause control circuit **922** to cause switch **938** to store the first running total of sampled time measurements as first sampled window **908** in a first memory such as first register **932** and to reset accumulator **940**.

(136) Upon sampling a given time measurement $t_{\text{sub.k}L}$ from clock signal input **906** and forwarded by skip circuit **950**, if counter **920** indicates that greater than (N-m)/L time measurements but less than (N-1)/L time measurements have been sampled, sampling circuit **902** may cause control circuit **922** to cause accumulator to add the sampled given time measurement to a second running total of sampled time measurements to generate second sampled window **910**.

(137) Upon sampling a given time measurement $t_{\text{sub.k}L}$ from clock signal input **906** and forwarded by skip circuit **950**, if counter **920** indicates that N/L or more time measurements have

been sampled, sampling circuit **902** may cause control circuit **922** to cause switch **938** to store the second running total of sampled time measurements as second sampled window **910** in a second memory such as second register **934**.

(138) Upon sampling of N/L time measurements from clock signal input **906**, estimation circuit **904** may be to apply first sampled window **908** and second sampled window **910** to adder **928** to subtract first sampled window **908** from second sampled window **910**, and to apply a result from adder **928** to multiplier **930** with a multiplication factor of $1/(m(N-m+L))$ to yield an estimated frequency or period of clock signal **912**.

(139) FIG. **12** is an illustration of a method **1200** for estimating a clock signal according to a least squares method, according to examples of the present disclosure. Method **1200** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **2-3**. Method **1200** may begin or end at any suitable point. Method **1200** may include more or fewer steps than shown, and the steps of method **1200** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1200**.

(140) At **1205**, it may be determined to estimate the frequency or period of a clock signal.

(141) At **1210**, N time measurements may be received on a clock signal input. A given time measurement may denote a respective time of a respective portion, such as a rising edge, of a given clock cycle of the clock signal input.

(142) **1215** and **1220** may be performed in parallel.

(143) At **1215**, a first sampled window may be generated from N time measurements. The first sampled window may include an accumulation of the N time measurements.

(144) At **1220**, a second sampled window may be generated from N time measurements. The second sampled window may include an accumulation of products. A given product may be a product of a given one of the N time measurements and a respective factor. The respective factor may be an index of the given time measurement.

(145) At **1225**, the frequency or the period of the clock signal may be estimated based on the first and second sampled windows.

(146) FIG. **13** is an illustration of a more detailed method **1300** for estimating a clock signal according to a least squares method, according to examples of the present disclosure. Method **1300** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **2-3**. Method **1300** may be a more detailed illustration of the method of FIG. **12**. Method **1300** may begin or end at any suitable point. Method **1300** may include more or fewer steps than shown, and the steps of method **1300** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1300**.

(147) At **1305**, it may be determined to estimate the frequency or period of a clock signal.

Accumulators and counters and switches may be reset. The counter may be to count an index of a number of received time measurements such that a first received time measurement is indexed at zero.

(148) At **1310**, a give time measurement of N time measurements may be received on a clock signal input. A given time measurement may denote a respective time of a respective portion, such as a rising edge, of a given clock cycle of the clock signal input.

(149) At **1315**, the counter may be incremented upon reception of the given time measurement to yield an index of received time measurements. The index may be used as a respective factor for given time measurement.

(150) **1320** and **1325** may be performed in parallel.

(151) At **1320**, the given time measurement may be optionally multiplied by 12, or this may be performed at **1335**. The given time measurement may be added to and accumulated in a running total of the N time measurements in a first accumulator in order to generate the first sampled

window for input into a least squares estimation.

(152) At **1325**, the given time measurement may be multiplied with a respective factor, such as the index of the given time measurement, to generate a given product. The given product may be added and accumulated into a running total of products in a second accumulator to generate a second sampled window for input into a least squares estimation.

(153) At **1330**, it may be determined whether the counter has reached $N-1$, indicating that N time measurements have been received. If so, method **1300** may proceed to **1335**. Otherwise, method **1300** may return to **1310**.

(154) At **1335**, the frequency or the period of the clock signal may be estimated based on the first and second sampled windows. A switch may be caused to output the values of the accumulators in the form of the sampled windows to an estimation circuit to make the estimation. The first sampled window may be multiplied by a least squares factor (12), if not done so in **1320**, to yield a first term. The second sampled window may be multiplied by a least squares factor ($6(N-1)$) to yield a second term. The multiplied first sampled window may be subtracted from the multiplied second sampled window to yield a third term. N may be applied to the results by dividing the third term by $(N(N\{ \text{circumflex over () } \}^2 - 1))$ to estimate frequency or period of the clock signal.

(155) FIG. **14** is an illustration of a method **1400** for estimating a clock signal using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure. Method **1400** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **4-7**. Method **1400** may begin or end at any suitable point. Method **1400** may include more or fewer steps than shown, and the steps of method **1400** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1400**.

(156) At **1405**, it may be determined to estimate the frequency or period of a clock signal.

(157) At **1410**, a first time measurement may be received on a clock signal input. A given time measurement denotes respective time of respective portion of given clock cycle of clock signal.

(158) At **1415**, a first sampled window may be generated from the first time measurement and a first previous time measurement received m time measurements earlier than the first time measurement.

(159) At **1420**, a second time measurement may be received on a clock signal input.

(160) At **1425**, a second sampled window may be generated from the second time measurement and a second previous time measurement received m time measurements earlier than the second time measurement.

(161) At **1430**, the frequency or the period of the clock signal may be estimated based on the first and second sampled windows.

(162) FIG. **15** is a more detailed illustration of a method **1500** for estimating a clock signal using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure. Method **1500** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **4-7**, specifically FIG. **5**. Method **1500** may be a more detailed illustration of the method of FIG. **14**. Method **1500** may begin or end at any suitable point. Method **1500** may include more or fewer steps than shown, and the steps of method **1500** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1500**.

(163) At **1505**, it may be determined to estimate the frequency or period of a clock signal.

Accumulators and counters and switches may be reset. The counter may be to count an index of a number of received time measurements such that a first received time measurement is indexed at zero.

(164) At **1510**, it may be determined whether N time measurements have been received. If so, method **1500** may proceed to **1535**. Otherwise, method **1500** may proceed to **1510** and **1530**, which may be performed in parallel.

(165) At **1510**, a given time measurement may be received on a clock signal input. The given time measurement may be provided to a delay circuit to delay given time measurement by m measurements. The value of m may be selected according to $N/3$ or $2N/3$.

(166) The given time measurement of N time measurements may be received on a clock signal input. A given time measurement may denote a respective time of a respective portion, such as a rising edge, of a given clock cycle of the clock signal input.

(167) At **1515**, a given previous time measurement may be retrieve from the delay circuit delayed by m measurements and received m time measurements earlier than the given time measurement.

(168) At **1520**, if the received time measurements are greater than m , then subtract the given previous time measurement from the given time measurement with an adder. Normalize result by dividing by m with a multiplier. Produce result as a given sampled window.

(169) At **1525**, a given sampled window may be stored in memory location such as first delay circuit of a chain of delay circuits. Method **1500** may return to **1505**.

(170) At **1530**, in a chain of delay circuits, a previously generated sampled window stored therein in a given delay circuit may be propagated to a subsequent, next successive delay circuit. Method **1500** may return to **1505**.

(171) **1535**, using a FIR filter with $N-m$ taps, the frequency or period of a clock signal may be estimated from at least two sampled windows of an available $N-m$ sampled windows stored in $N-m$ delay circuits. The sampled windows may be added together and divided by the number of sampled windows such as $(N-m)$.

(172) FIG. **16** is an illustration of yet another more detailed method **1600** for estimating a clock signal using comparisons of time measurements with previously made time measurements, according to examples of the present disclosure. Method **1600** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **4-7**, specifically FIG. **6**. Method **1600** may be a more detailed illustration of the method of FIG. **14** or FIG. **15**. Method **1600** may begin or end at any suitable point. Method **1600** may include more or fewer steps than shown, and the steps of method **1600** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1600**.

(173) At **1605**, it may be determined to estimate the frequency or period of a clock signal. Accumulators and counters and switches may be reset. The counter may be to count an index of a number of received time measurements such that a first received time measurement is indexed at zero.

(174) At **1610**, it may be determined whether N time measurements have been received. If so, method **1600** may proceed to **1630**. Otherwise, method **1600** may proceed to **1615**.

(175) At **1615**, it may be determined whether more than m time measurements have been received. If so, method **1600** may proceed to **1620**. Otherwise, method **1600** may proceed to **1610**.

(176) At **1620**, a given time measurement on a clock signal input of a quantity $N-m$ successive time measurements may be received. A frequency or period for the clock signal for a given partial time period may be estimated. The partial time period may be the given time measurement less the delayed time measurement. The delayed time measurement may be delayed by m measurements with respect to given time measurement. The period for the given partial time period may be the partial time period divided by m .

(177) At **1625**, the estimation of the frequency or period for the clock signal for the partial time period may be stored as a given sampled window. Method **1600** may return to **1610**.

(178) At **1630**, the stored estimations of the frequency or period of the clock signal for the given partial time periods may be averaged to yield the estimated frequency or period of the clock signal.

(179) FIG. **17** is an illustration of a method **1700** for estimating a clock signal, according to examples of the present disclosure. Method **1700** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **8-11**. Method **1700** may begin or end at any

suitable point. Method **1700** may include more or fewer steps than shown, and the steps of method **1700** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1700**.

(180) At **1705**, it may be determined to estimate the frequency or period of a clock signal.

(181) At **1710**, receive N time measurements on a clock signal input. A given time measurement may denote a respective time of respective portion of given clock cycle of clock signal.

(182) At **1715**, a first sampled window may be generated from the N time measurements, including a sum of two or more of the first m of the N time measurements.

(183) At **1720**, a second sampled window may be generated from the N time measurements, including a sum of two or more of the last m of the N time measurements.

(184) At **1725**, the frequency or the period of the clock signal may be estimated based on the first and second sampled windows.

(185) FIG. **18** is an illustration of a method **1800** for estimating a clock signal, according to examples of the present disclosure. Method **1800** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **8-11**, specifically FIGS. **9-10**. Method **1800** may be a more detailed illustration of the method of FIG. **17**. Method **1800** may begin or end at any suitable point. Method **1800** may include more or fewer steps than shown, and the steps of method **1800** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1800**.

(186) At **1805**, it may be determined to estimate the frequency or period of a clock signal. Accumulators and counters and switches may be reset. The counter may be to count an index of a number of received time measurements such that a first received time measurement is indexed at zero.

(187) At **1810**, a given time measurement may be received on a clock signal input. A given time measurement may denote respective time of respective portion of given clock cycle of clock signal.

(188) At **1815**, it may be determined whether less than m time measurements have been received. If so, method **1800** may proceed to **1835**. Otherwise, method **1800** may proceed to **1820**.

(189) At **1820**, a first running total of time measurements may be maintained in a first accumulator.

(190) At **1825**, it may be determined whether the number of received measurements is greater than or equal to N-m but less than or equal to N. If so, method **1800** may proceed to **1840**. Otherwise, method **1800** may proceed to **1830**.

(191) At **1830**, it may be determined whether N time measurements have been received. If so, method **1800** may proceed to **1845**. Otherwise, method **1800** may proceed to **1810**.

(192) At **1835**, a given time measurement may be added to a first accumulator to add the given time measurement to a first running total of time measurements to generate first sampled window. The first accumulator may be used to add successive time measurements of the first m time measurements. Method **1800** may proceed to **1825**.

(193) At **1840**, a given time measurement may be added to a second accumulator to add the given time measurement to a second running total of time measurements to generate second sampled window. The second accumulator may be used to add successive time measurements of the last m time measurements. Method **1800** may proceed to **1830**.

(194) At **1845**, the first sampled window may be subtracted from the second sampled window, and divide by $(m(N-m))$ to yield the estimated frequency or period of the clock signal.

(195) In FIG. **18**, if m is less than $N/2$, a single accumulator may be used, wherein the references to the first accumulator and the second accumulator may be a same accumulator.

(196) FIGS. **19A** and **19B** are an illustration of a method **1900** for estimating a clock signal, according to examples of the present disclosure. Method **1900** may illustrate operation of or be implemented by any suitable apparatus, such as the elements of FIGS. **8-11**, specifically FIG. **11**. Method **1900** may be a more detailed illustration of the method of FIG. **17** or FIG. **18**. Method **1900** may begin or end at any suitable point. Method **1800** may include more or fewer steps than

shown, and the steps of method **1900** may be optionally repeated, omitted, performed in a different order, performed recursively or in parallel with one another, or recursively or in parallel with another instance of method **1900**.

(197) At **1905**, it may be determined to estimate the frequency or period of a clock signal. Accumulators and counters and switches may be reset. The counter may count an index of a number of sampled and received time measurements such that a first sampled time measurement and a first received time measurement are indexed at zero.

(198) At **1910**, a given time measurement may be received on a clock signal input. A given time measurement may denote respective time of respective portion of given clock cycle of clock signal.

(199) At **1920**, it may be determined whether to sample given time measurement. Samples may be skipped for L received time measurements after a last sampled time measurement so as to sample N/L time measurements over the course of N received time measurements. If samples are to be skipped, method **1900** may proceed to **1910**. Otherwise, method **1900** may proceed to **1925**.

(200) At **1925**, the time measurement may be sampled. It may be determined whether less than m/L time measurements have been sampled. If so, method **1900** may proceed to **1935**. Otherwise, method **1900** may proceed to **1930**.

(201) At **1930**, a first running total of sampled time measurements may be stored as a first sampled window in a first memory such as a first register and to reset the accumulator.

(202) At **1935**, the sampled given time measurement may be added to the accumulator to add it to a first running total of sampled time measurements to generate the first sampled window.

(203) At **1940**, it may be determined whether more than $(N-m)/L$ samples have been taken but less than $(N-1)/L$. If so, method **1900** may proceed to **1950**. Otherwise, method **1900** may proceed to **1945**.

(204) At **1945**, a second running total of sampled time measurements may be stored as a second sampled window in a first memory such as a second register and to reset the accumulator.

(205) At **1950**, the sampled given time measurement may be added to the accumulator to add it to a second running total of sampled time measurements to generate the second sampled window.

(206) At **1955**, it may be determined whether N/L time measurements have been sampled. If so, method **1900** may proceed to **1960**. Otherwise, method **1900** may proceed to **1910**.

(207) At **1960**, the first sampled window may be subtracted from the second sampled window, and divide by $(m(N-m+L))$ to yield the estimated frequency or period of the clock signal.

(208) Examples of the present disclosure may include an apparatus. The apparatus may include a clock signal input to receive N time measurements to estimate a frequency or period of a clock signal. A time measurement of the N time measurements may be to denote a respective time of a respective portion of a given cycle of a clock signal. The clock signal may include a frequency or period to be estimated over the N time measurements.

(209) The apparatus may include a sampling circuit. The sampling circuit may be to, upon reception of a first time measurement on the clock signal input, generate a first sampled window from the clock signal input. The first sampled window may be based upon the first time measurement and a first previous time measurement. The first previous time measurement may be a time measurement which was received m time measurements earlier than the first time measurement, wherein m is greater than one. The sampling circuit may, upon reception of a second time measurement on the clock signal input, generate a second sampled window from the clock signal input. The second sampled window may be based upon the second time measurement and a second previous time measurement. The second previous time measurement a time measurement may be received m time measurements earlier than the second time measurement. The second time measurement may be subsequent to the first time measurement.

(210) The estimation circuit may estimate the frequency or period of the clock signal based upon the first sampled window and the second sampled window.

(211) In combination with any of the above examples, the sampling circuit may include a delay

circuit to delay the first previous time measurement by m measurements.

(212) In combination with any of the above examples, the sampling circuit may be to, upon reception of the first time measurement, subtract the first previous time measurement from the first time measurement to obtain the first sampled window, and store the first sampled window in a first memory location. Upon reception of the second time measurement, the sampling circuit may subtract the second previous time measurement from the second time measurement to obtain the second sampled window and store the second sampled window in a second memory location.

(213) In combination with any of the above examples, the sampling circuit may divide a value of subtracting the first previous time measurement from the first time measurement by a number of delayed measurements m to obtain the first sampled window.

(214) In combination with any of the above examples, the first memory location may be a delay circuit in a chain of delay circuits. The delay circuit may propagate its value to a subsequent delay circuit in the chain of delay circuits upon a reception of a given time measurement on the clock signal input.

(215) In combination with any of the above examples, the second memory location may be the delay circuit in the chain of delay circuits. The sampling circuit may store the first sampled window in the delay circuit upon reception of the first time measurement on the clock signal input. The delay circuit may propagate the first sampled window to a next delay circuit upon reception of the second time measurement on the clock signal input. The sampling circuit may store the second sampled window in the delay circuit upon reception of the second time measurement on the clock signal input.

(216) In combination with any of the above examples, the sampling circuit may generate a quantity $(N-m)$ sampled windows to make the estimation.

(217) In combination with any of the above examples, the estimation circuit includes a finite impulse response (FIR) filter to estimate, from at least the first sampled window and the second sampled window, the frequency or period of the clock signal.

(218) In combination with any of the above examples, the FIR filter may include a quantity of $(N-m)$ taps.

(219) In combination with any of the above examples, the sampling circuit may generate a given sampled window upon reception of a given time measurement on the clock signal input of a quantity $(N-m)$ successive time measurements received on the clock signal input. The given sampled window may include the first sampled window and the second sampled window. The sampling circuit may store the given sampled window in a delay circuit of a chain of delay circuits. Upon reception the given time measurement on the clock signal input of a quantity $(N-m)$ successive time measurements received on the clock signal input, the delay circuit may propagate a previously given sampled window to a next successive delay circuit in the chain of delay circuits. After reception of N successive time measurements on the clock signal input, the estimation circuit may use $(N-m)$ sampled windows from $(N-m)$ delay circuits in the chain of delay circuits to estimate the frequency or period of the clock signal.

(220) In combination with any of the above examples, upon reception of a given time measurement on the clock signal input of a quantity $(N-m)$ successive time measurements received on the clock signal input, the sampling circuit may generate an estimation of a frequency or period of the clock signal for a given partial time period. The given partial time period between the given time measurement and a delayed time measurement may be delayed by m measurements with respect to the given time measurement. The sampling circuit may store the estimation of the frequency or period of the clock signal for the given partial time period. Upon reception of a quantity N successive time measurements on the clock signal input, the estimation circuit may average stored estimations of the frequency or period of the clock signal for the given partial time periods.

(221) In combination with any of the above examples, m may be selected according to $N/3$ or $2N/3$.

(222) Alone or in combination with any of the above examples, examples of the present disclosure

may include an apparatus with a clock signal input to receive N time measurements to estimate a frequency or period of a clock signal. A time measurement of the N time measurements may be to denote a respective time of a respective portion of a given cycle of a clock signal. The clock signal may include a frequency or period to be estimated over the N time measurements.

(223) The apparatus may include a sampling circuit to generate a first sampled window from the clock signal input. The first sampled window may include a sum of a plurality of a first m of the N time measurements. The sampling circuit may generate a second sampled window from the clock signal input. The second sampled window may include a sum of a plurality of a last m of the N time measurements. The apparatus may include an estimation circuit to estimate the frequency or period of the clock signal based upon the first sampled window and the second sampled window.

(224) In combination with any of the above examples, the sampling circuit may include a first accumulator to generate the first sampled window and a second accumulator to generate the second sampled window.

(225) In combination with any of the above examples, m may be greater than $N/2$.

(226) In combination with any of the above examples, the apparatus may include a counter to count time measurements as they are received on the clock signal input. Upon reception of a given time measurement on the clock signal input, if less than m time measurements have been received, the first accumulator may add the given time measurement to a first running total of time measurements to generate the first sampled window. Upon reception of a given time measurement on the clock signal input, if m or greater time measurements have been received, the first accumulator may maintain the first running total of time measurements. Upon reception of a given time measurement on the clock signal input, if $(N-m)$ or greater time measurements and less than or equal to N time measurements have been received, the second accumulator may add the given time measurement to a second running total of time measurements to generate the second sampled window.

(227) In combination with any of the above examples, the sampling circuit may include an accumulator to add successive time measurements of the first m of the N time measurements to generate the first sampled window.

(228) In combination with any of the above examples, m is less than $N/2$ and the sampling circuit may include an accumulator to first generate the first sampled window and to then generate the second sampled window.

(229) In combination with any of the above examples, the apparatus may include a counter to count time measurements as they are received on the clock signal input, wherein, upon reception of a given time measurement on the clock signal input, if the counter indicates that less than m time measurements have been received, an accumulator may add the given time measurement to a first running total of time measurements to generate the first sampled window. Upon reception of a given time measurement on the clock signal input, if the counter indicates that m time measurements have been received, the sampling circuit may store the first running total of time measurements as the first sampled window in a first memory and to reset the accumulator. Upon reception of a given time measurement on the clock signal input, if the counter indicates that a quantity of time measurements greater than or equal to $(N-m)$ but less than N has been received, the accumulator may add the given time measurement to a second running total of time measurements. Upon reception of a given time measurement on the clock signal input, if the counter indicates that N time measurements have been received, the sampling circuit may store the second running total of time measurements as the second sampled window in a second memory.

(230) In combination with any of the above examples, the accumulator may accumulate the first m time measurements and store an accumulation of the first m time measurements in a first memory as the first sampled window, and accumulate the last m time measurements and store an accumulation of the last m time measurements in a second memory as the second sampled window.

(231) In combination with any of the above examples, the estimation circuit may, upon the

sampling circuit reception of N time measurements, subtract the first sampled window from the second sampled window, and apply a result from the subtraction to a multiplication factor of $1/(m(N-m))$ to yield an estimated frequency or period of the clock signal.

(232) In combination with any of the above examples, the sampling circuit may sample a given time measurement and to skip sampling of a next L time measurements after the given time measurement and to exclude the next L time measurements from generation of the first sampled window and the second sampled window.

(233) In combination with any of the above examples, the sampling circuit may sample N/L time measurements of the N time measurements.

(234) In combination with any of the above examples, the sampling circuit may accumulate a first quantity (m/L) of sampled time measurements of the N time measurements and store the first quantity (m/L) of sampled time measurements in a first memory as the first sampled window. The sampling circuit may accumulate a last quantity (m/L) of sampled time measurements of the N time measurements and store the last quantity (m/L) of sampled time measurements in a second memory as the second sampled window.

(235) In combination with any of the above examples, the apparatus may include an accumulator to first generate the first sampled window and to then generate the second sampled window. The apparatus may include a counter to count sampled time measurements as they are sampled by the sampling circuit. The sampling circuit may, upon sampling a given time measurement on the clock signal input, if the counter indicates that less than m/L time measurements have been sampled, use the accumulator to add the sampled given time measurement to a first running total of sampled time measurements to generate the first sampled window. Upon sampling a given time measurement on the clock signal input, once the counter indicates that m/L or more time measurements have been sampled, the sampling circuit may store the first running total of sampled time measurements as the first sampled window in a first memory and to reset the accumulator. Upon sampling a given time measurement on the clock signal input, if the counter indicates that greater than $(N-m)/L$ time measurements but less than $(N-1)/L$ time measurements have been sampled, the accumulator may add the sampled given time measurement to a second running total of sampled time measurements. Upon sampling a given time measurement on the clock signal input, if the counter indicates that N/L or more time measurements have been sampled, the sampling circuit may store the second running total of sampled time measurements as the second sampled window in a second memory.

(236) In combination with any of the above examples, the estimation circuit may, upon the sampling circuit sampling N/L or more time measurements, subtract the first sampled window from the second sampled window, and to apply a result to a multiplication factor of $1/(m(N-m+L))$ to yield an estimated frequency or period of the clock signal.

(237) Alone or in combination with any of the above examples, examples of the present disclosure may include an apparatus with a clock signal input to receive N time measurements to estimate a frequency or period of a clock signal. A time measurement of the N time measurements may be to denote a respective time of a respective portion of a given cycle of a clock signal. The clock signal may include a frequency or period to be estimated over the N time measurements.

(238) The apparatus may include a sampling circuit to generate a first sampled window from the clock signal input. The first sampled window may include an accumulation of the N time measurements. The sampling circuit may generate a second sampled window from the clock signal input. The second sampled window may include an accumulation of a plurality of products, wherein a given product is a product of one of the N time measurements and a respective factor. The apparatus may include an estimation circuit to estimate the frequency or period of the clock signal based upon the first sampled window and the second sampled window.

(239) In combination with any of the above examples, the sampling circuit may generate the first sampled window and the second sampled window for input into a least squares estimation by the estimation circuit.

- (240) In combination with any of the above examples, the sampling circuit may, upon reception of a given time measurement on the clock signal input, accumulate the given time measurement into a running total of the N time measurements in a first accumulator to generate the first sampled window.
- (241) In combination with any of the above examples, the sampling circuit may, upon reception of a given time measurement on the clock signal input, increment an index of received time measurements to generate the respective factor, multiply the given time measurement with the index to generate a given product, and accumulate the given product into a running total of products in a second accumulator to generate the second sampled window.
- (242) In combination with any of the above examples, the sampling circuit may, upon reception of N time measurements, multiply the second sampled window by a least squares factor and provide the first sampled window and the second sampled window to the estimation circuit.
- (243) In combination with any of the above examples, estimation circuit may, upon reception of the first sampled window and the second sampled window, subtract the first sampled window from the second sampled window to yield a result and apply N to the result to estimate the frequency or period of the clock signal.
- (244) In combination with any of the above examples, the apparatus may include a counter, a comparator, a first accumulator, a second accumulator, and a switch. The counter may count an index of a number of received time measurements, a first received time measurement indexed at zero. The comparator may evaluate whether the index of the number of received time measurements has reached N-1 and to provide the evaluation to the switch. For a given received time measurement, the sampling circuit may multiply the index with the received time measurement and store a result in the first accumulator. For a given received time measurement, the sampling circuit may accumulate the received time measurement in the second accumulator. Upon reaching N time measurements, the switch may cause an output of the first accumulator and an output of the second accumulator to be provided to the estimation circuit. Upon reaching N time measurements, the estimation circuit may multiply the output of the first accumulator by 12 to yield a first term, multiply the output of the second accumulator by $6(N-1)$ to yield a second term, subtract the first term from the second term to yield a third term, and divide the third term by $(N(N^2-1))$ to estimate the frequency or period of the clock signal.
- (245) In combination with any of the above examples, the apparatus may include a counter, a comparator, a first accumulator, a second accumulator, and a switch. The counter may count an index of a number of received time measurements. A first received time measurement may be indexed at zero. The comparator may evaluate whether the index of the number of received time measurements has reached N-1 and to provide the evaluation to the switch. For a given received time measurement, the sampling circuit may multiply the index with the received time measurement and by 12 and store a result in the first accumulator. For a given received time measurement, the sampling circuit may accumulate the received time measurement in the second accumulator. Upon reaching N time measurements, the switch may cause an output of the first accumulator and an output of the second accumulator to be provided to the estimation circuit. Upon reaching N time measurements, the estimation circuit may multiply the output of the second accumulator by $6(N-1)$ to yield a first term, subtract the output of the first accumulator from the first term to yield a second term, and divide the second term by $(N(N^2-1))$ to estimate the frequency or period of the clock signal.
- (246) Although example embodiments have been described above, other variations and embodiments may be made from this disclosure without departing from the spirit and scope of these embodiments.

Claims

1. An apparatus, comprising: a clock signal input to receive N time measurements, a time measurement of the N time measurements to denote a respective time of a respective portion of a given cycle of a clock signal, the clock signal to include a frequency or period to be estimated over the N time measurements; a sampling circuit to: generate a first sampled window from the clock signal input, the first sampled window including an accumulation of the N time measurements; and generate a second sampled window from the clock signal input, the second sampled window including an accumulation of a plurality of products, wherein a given product is a product of one of the N time measurements and a respective factor; and an estimation circuit to estimate the frequency or period of the clock signal based upon the first sampled window and the second sampled window.
2. The apparatus of claim 1, wherein the sampling circuit is to generate the first sampled window and the second sampled window for input into a least squares estimation by the estimation circuit.
3. The apparatus of claim 1, wherein the sampling circuit is to, upon reception of a given time measurement on the clock signal input, accumulate the given time measurement into a running total of the N time measurements in a first accumulator to generate the first sampled window.
4. The apparatus of claim 1, wherein the sampling circuit is to, upon reception of a given time measurement on the clock signal input: increment an index of received time measurements to generate the respective factor; multiply the given time measurement with the index to generate a given product; and accumulate the given product into a running total of products in a second accumulator to generate the second sampled window.
5. The apparatus of claim 1, wherein the sampling circuit is to, upon reception of N time measurements, multiply the second sampled window by a least squares factor and provide the first sampled window and the second sampled window to the estimation circuit.
6. The apparatus of claim 1, wherein the estimation circuit is to, upon reception of the first sampled window and the second sampled window, subtract the first sampled window from the second sampled window to yield a result and apply N to the result to estimate the frequency or period of the clock signal.
7. The apparatus of claim 1, comprising a counter, a comparator, a first accumulator, a second accumulator, and a switch, wherein: the counter is to count an index of a number of received time measurements, a first received time measurement indexed at zero; the comparator is to evaluate whether the index of the number of received time measurements has reached N-1 and to provide the evaluation to the switch; for a given received time measurement, the sampling circuit is to multiply the index with the received time measurement and store a result in the first accumulator; for a given received time measurement, the sampling circuit is to accumulate the received time measurement in the second accumulator; upon reaching N time measurements, the switch is to cause an output of the first accumulator and an output of the second accumulator to be provided to the estimation circuit; upon reaching N time measurements, the estimation circuit is to: multiply the output of the first accumulator by 12 to yield a first term; multiply the output of the second accumulator by $6(N-1)$ to yield a second term; subtract the first term from the second term to yield a third term; and divide the third term by $(N(N-1))$ to estimate the frequency or period of the clock signal.
8. The apparatus of claim 1, comprising a counter, a comparator, a first accumulator, a second accumulator, and a switch, wherein: the counter is to count an index of a number of received time measurements, a first received time measurement indexed at zero; the comparator is to evaluate whether the index of the number of received time measurements has reached N-1 and to provide the evaluation to the switch; for a given received time measurement, the sampling circuit is to multiply the index with the received time measurement and by 12 and store a result in the first accumulator; for a given received time measurement, the sampling circuit is to accumulate the received time measurement in the second accumulator; upon reaching N time measurements, the

switch is to cause an output of the first accumulator and an output of the second accumulator to be provided to the estimation circuit; upon reaching N time measurements, the estimation circuit is to: multiply the output of the second accumulator by $6(N-1)$ to yield a first term; subtract the output of the first accumulator from the first term to yield a second term; and divide the second term by $(N(N\{circumflex\over{ } \}^2-1))$ to estimate the frequency or period of the clock signal.

9. A method, comprising: receiving N time measurements on a clock signal input, a time measurement of the N time measurements to denote a respective time of a respective portion of a given cycle of a clock signal, the clock signal to include a frequency or period to be estimated over the N time measurements; generating a first sampled window from the clock signal input, the first sampled window including an accumulation of the N time measurements; generating a second sampled window from the clock signal input, the second sampled window including an accumulation of a plurality of products, wherein a given product is a product of one of the N time measurements and a respective factor; and estimating the frequency or period of the clock signal based upon the first sampled window and the second sampled window.

10. The method of claim 9, comprising generating the first sampled window and the second sampled window for input into a least squares estimation by the estimation circuit.

11. The method of claim 9, comprising, upon reception of a given time measurement on the clock signal input, accumulating the given time measurement into a running total of the N time measurements in a first accumulator to generate the first sampled window.

12. The method of claim 9, comprising, upon reception of a given time measurement on the clock signal input: incrementing an index of received time measurements to generate the respective factor; multiplying the given time measurement with the index to generate a given product; and accumulating the given product into a running total of products in a second accumulator to generate the second sampled window.

13. The method of claim 9, comprising, upon reception of N time measurements, multiplying the second sampled window by a least squares factor and provide the first sampled window and the second sampled window to the estimation circuit.

14. The method of claim 9, comprising, upon reception of the first sampled window and the second sampled window, subtracting the first sampled window from the second sampled window to yield a result and apply N to the result to estimate the frequency or period of the clock signal.

15. The method of claim 9, comprising: operating a counter to count an index of a number of received time measurements, a first received time measurement indexed at zero; operating a comparator to evaluate whether the index of the number of received time measurements has reached N-1 and to provide the evaluation to a switch; for a given received time measurement, multiplying the index with the received time measurement and store a result in a first accumulator; for a given received time measurement, accumulating the received time measurement in a second accumulator; upon reaching N time measurements, operating the switch is to cause an output of the first accumulator and an output of the second accumulator to be provided for estimation; and upon reaching N time measurements: multiplying the output of the first accumulator by 12 to yield a first term; multiplying the output of the second accumulator by $6(N-1)$ to yield a second term; subtracting the first term from the second term to yield a third term; and dividing the third term by $(N(N\{circumflex\over{ } \}^2-1))$ to estimate the frequency or period of the clock signal.

16. The method of claim 9, comprising: operating a counter to count an index of a number of received time measurements, a first received time measurement indexed at zero; operating a comparator to evaluate whether the index of the number of received time measurements has reached N-1 and to provide the evaluation to a switch; for a given received time measurement, multiplying the index with the received time measurement and by 12 and storing a result in a first accumulator; for a given received time measurement, accumulating the received time measurement in a second accumulator; upon reaching N time measurements, operating the switch is to cause an output of the first accumulator and an output of the second accumulator to be provided for

estimation; and upon reaching N time measurements: multiplying the output of the second accumulator by $6(N-1)$ to yield a first term; subtracting the output of the first accumulator from the first term to yield a second term; and dividing the second term by $(N(N\{\text{circumflex over } (\quad)\}^2-1))$ to estimate the frequency or period of the clock signal.
